PC COMPONENTS CHEAT SHEET

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Note: The brands, models, and specifications listed below are examples of current mainstream options. Many other manufacturers, form factors, and configurations exist in the market.

🖥️ MOTHERBOARD

Function: Main circuit board connecting all components

- **Form Factors:** ATX, Micro-ATX, Mini-ITX
- **Key Features:** CPU socket, RAM slots, expansion slots, I/O ports
- **Common Sockets:** Intel (LGA 1700, LGA 1200) | AMD (AM4, AM5)
- **Chipset:** Controls data flow between components

🧠 CPU (Processor)

Function: Executes instructions and performs calculations

- **Key Specs:** Clock speed (GHz), cores, threads, cache
- **Major Brands:** Intel (Core i3/i5/i7/i9) | AMD (Ryzen 3/5/7/9)
- **Architecture:** x86-64, instruction sets (SSE, AVX)
- **Cooling Required:** Air cooler or liquid cooling

💾 RAM (Memory)

Function: Temporary storage for active programs and data

- **Types:** DDR4, DDR5 (newer and faster)
- **Speeds:** Measured in MHz (e.g., 3200MHz, 3600MHz)
- **Capacity:** 8GB, 16GB, 32GB, 64GB per stick
- **Channels:** Dual-channel (2 sticks) for better performance

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💿 STORAGE DEVICES

SSD (Solid State Drive)

- **Faster, no moving parts, more expensive**
- **Types:** SATA SSD, M.2 NVMe SSD

- **Speeds:** SATA (up to 550MB/s) | NVMe (up to 7000MB/s)

HDD (Hard Disk Drive)

- **Slower, mechanical, cheaper per GB**
- **Speeds:** 5400 RPM, 7200 RPM
- **Best for:** Mass storage, backups

🎮 GPU (Graphics Card)

Function: Renders images, videos, and handles graphics processing

- **Types:** Integrated (built into CPU) | Dedicated (separate card)
- **Major Brands:** NVIDIA (GeForce RTX/GTX) | AMD (Radeon RX)
- **Memory:** VRAM (4GB, 8GB, 12GB, 16GB+)
- **Uses:** Gaming, video editing, AI/machine learning

⚡ PSU (Power Supply Unit)

Function: Converts AC power to DC power for components

- **Wattage:** 450W, 650W, 750W, 850W+ (choose based on components)
- **Efficiency:** 80+ Bronze, Silver, Gold, Platinum, Titanium
- **Modular:** Cables can be removed for better airflow
- **Connectors:** 24-pin ATX, 8-pin CPU, PCIe, SATA

🌬️ COOLING SYSTEMS

CPU Coolers

- **Air Coolers:** Heatsink + fan (quieter, cheaper)
- **Liquid Coolers:** AIO (All-in-One) or custom loops

Case Fans

- **Intake:** Front/bottom fans bring cool air in
- **Exhaust:** Rear/top fans push hot air out
- **Sizes:** 120mm, 140mm common

🏠 CASE

Function: Houses and protects all components

- **Form Factors:** Full Tower, Mid Tower, Mini-ITX
- **Features:** Cable management, airflow, USB ports
- **Materials:** Steel, aluminum, tempered glass panels

🔌 COMMON CONNECTORS & CABLES

Power Connectors

- **24-pin ATX:** Main motherboard power
- **8-pin CPU:** CPU power (top-left of motherboard)
- **6+2 pin PCIe:** Graphics card power
- **SATA Power:** Storage drives

Data Cables

- **SATA:** HDD/SSD to motherboard
- **M.2:** Direct motherboard connection for NVMe SSDs
- **USB:** Front panel connectors

This cheat sheet covers the essential PC components every IT student should know. Keep this handy for building, upgrading, or troubleshooting computers!